Lecture 1.8: Strings in JavaScript

What is a string?

A **string** is a sequence of characters (text). JavaScript provides many built-in methods to work with strings.

Starter text (from your lecture)

```
var txt = "a quick brown fox jumps over the lazy dog";
```

Output:

```
No console output (declaration)
```

Length and searching

`.length`

```
console.log(txt.length);
```

Output:

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```
41
```

`indexOf`

`indexOf` returns the position of the first match or `-1` if not found:

```
console.log(txt.indexOf("o"));
console.log(txt.indexOf("zebra")); // -1
```

Output:

```
10
-1
```

Case conversion

```
console.log(txt.toLowerCase());
console.log(txt.toUpperCase());
```

Output:

```
a quick brown fox jumps over the lazy dog
A QUICK BROWN FOX JUMPS OVER THE LAZY DOG
```


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Extracting parts of a string

Method	Usage	Notes
`slice(start, end)`	`txt.slice(2, 15)`	End is exclusive; supports negative indexes
`substring(start, end)`	`txt.substring(2, 15)`	End is exclusive; does NOT support negative indexes
`substr(start, length)`	`txt.substr(2, 10)`	Older; works but not preferred in new code

Examples

```
console.log(txt.slice(2, 15));      // characters 2..14
console.log(txt.substr(2, 10));     // 10 characters from index 2
console.log(txt.substring(2, 15));  // similar to slice (no negatives)
```

Output:

```
quick brown f
quick brow
quick brown f
```

Replacing text

`replace` replaces the first match by default:

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```
console.log(txt.replace("fox", "cat"));
```

Output:

```
a quick brown cat jumps over the lazy dog
```

> **Extra insight:** If you need to replace all occurrences, you can use `replaceAll` (modern browsers) or a regular expression.

```
// Replace all spaces with hyphens
console.log(txt.replaceAll(" ", "-"));
```

Output:

```
a-quick-brown-fox-jumps-over-the-lazy-dog
```

Characters and character codes

```
console.log(txt.charAt(5));
console.log(txt.charCodeAt(5));
```

Output:

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```
c
99
```

Splitting and trimming

`split`

```
var words = txt.split(" ");
console.log(words);
```

Output:

```
["a","quick","brown","fox","jumps","over","the","lazy","dog"]
```

`trim`

```
var messy = "   hello   ";
console.log(messy.trim()); // "hello"
```

Output:

```
hello
```


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Checking contents

```
console.log(txt.includes("lazy"));
console.log(txt.startsWith("a quick"));
console.log(txt.endsWith("dog"));
```

Output:

```
true
true
true
```

Mini task: word counter

Count how many words are in a sentence:

```
function wordCount(sentence) {
    if (sentence === null || sentence === undefined) return 0;

    var s = sentence.trim();
    if (s === "") return 0;

    return s.split(/\s+/).length;
}

console.log(wordCount("a quick brown fox")); // 4
```

Output:

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```
4
```

Practice tasks

1. Ask the user for a sentence and print its length.
2. Print the first and last character of the sentence.
3. Replace a word in the sentence and print the new result.
4. Split the sentence into words and print the array.

JS Refined Notes — Lecture 1.8

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